

CARDIAC CYCLE

Sequence of Events: Order in which the heart contracts and relaxes

Duration: Time taken for one complete cycle

Heart Rate (HR): Number of heartbeats per minute

TYPES OF CYCLES

Atrial Cycle: Events occurring in the atria

Ventricular Cycle: Events occurring in the ventricles

PHASES OF CARDIAC CYCLE

Systole: Contraction phase

Diastole: Relaxation phase

ATRIAL CYCLE

Atrial Systole (AS): Contraction of the atria to push blood into the ventricles

Dynamic Phase: Initial active contraction of the atria

Adynamic Phase: Later phase with no active contraction

Pressure Gradient: Difference in pressure that allows blood to flow

Atrial Diastole (AD): Relaxation of the atria

A-V Valves: Atrioventricular valves (Mitral and Tricuspid) that separate the atria from the ventricles

VENTRICULAR CYCLE

Ventricular Systole (VS): Contraction of ventricles to pump blood into arteries

First Heart Sound: "Lub" sound due to A-V valve closure

PHASES OF VENTRICULAR SYSTOLE

Isovolumetric Contraction: No blood movement, only pressure buildup

Semilunar Valves: Valves between the ventricles and arteries (Aortic & Pulmonary)

Maximum Ejection: Peak blood flow from ventricles into arteries

Reduced Ejection: Slower blood flow due to reduced pressure

Second Heart Sound: "Dub" sound due to semilunar valve closure

VENTRICULAR DIASTOLE (VD)

Protodiastole: Early diastole phase where blood flow slows down

Isovolumetric Relaxation: No blood movement, only pressure drop

Ventricular Filling: Blood flows from atria into ventricles

PHASES OF VENTRICULAR FILLING

First Rapid Filling: Fastest blood flow into ventricles

Third Heart Sound: Sound produced during rapid filling

Slow Filling: Gradual blood flow

Second Rapid Filling: Final phase of filling

Fourth Heart Sound: Sound during final filling phase

OTHER EVENTS IN THE CARDIAC CYCLE

Pressure Changes: Fluctuations in heart and blood vessel pressure

Blood Volume Changes: Variation in blood content within heart chambers

Arterial & Venous Pulses: Waves in blood vessels due to heartbeat

EKG (Electrocardiogram) Generation: Electrical signals recorded from the heart

Heart Sounds Production: Sounds due to valve actions

Apical Impulse (Apex Beat): The heartbeat felt at the chest wall

FACTORS AFFECTING CARDIAC CYCLE

Heart Rate: Number of beats per minute

Duration: Time taken for one complete cycle

Contractility: Strength of heart contraction

Systole: Contraction phase of the heart

Preload: Stretching of heart muscle before contraction

Venous Return: Blood returning to the heart

Stroke Volume: Amount of blood pumped per beat

Afterload: Resistance the heart overcomes to eject blood

Ejection Phase: Period when blood leaves the ventricles

Autonomic Nervous System: Controls involuntary heart functions

Sympathetic Stimulation: Increases heart rate and contraction strength

Parasympathetic Stimulation: Slows heart rate

Blood Volume: Total amount of blood in circulation

Age: Affects heart rate and cycle duration

Temperature: Body heat influencing heart rate

Electrolyte Balance: Proper levels of minerals affecting heart function

Potassium & Calcium: Essential minerals for heart activity

PROPERTIES OF CARDIAC MUSCLES

Contractility: Ability to contract and pump blood

Excitability: Ability to respond to electrical impulses

Rhythmicity: Regular and repetitive contractions

Automaticity: Ability to generate electrical impulses independently

Conductivity: Ability to transmit electrical signals across muscle fibers

Refractoriness: Period after contraction when no new impulse can trigger a contraction

Tetany: Sustained muscle contraction, which the heart avoids

Elasticity: Ability to return to original shape after stretching

Endurance: Ability to keep working without fatigue

ELECTROCARDIOGRAM (ECG)

Graphical Recording: A visual representation of heart's electrical activity

Summated Electrical Activity: Combined electrical signals from different heart regions

PQRST COMPLEX

Complex of Waves: A set of deflections representing heart activity

Isoelectric Line: The baseline indicating no electrical activity

WAVE CHARACTERISTICS

P Wave: Represents **atrial depolarization** (electrical activation of atria)

Depolarization: A change in electrical charge causing muscle contraction

Voltage: Electrical potential difference

Q Wave: Represents **septal depolarization** (activation of the septum)

Septum: The wall dividing the heart's ventricles

R Wave: Electrical potential moving toward the **apex** (bottom tip of the heart)

Cardioversion: A medical procedure to restore normal heart rhythm

S Wave: Electrical potential moving toward the **base** (top part of the heart)

T Wave: Represents **ventricular repolarization** (recovery phase of ventricles)

Repolarization: Restoration of electrical charge to resting state

U Wave: Due to delayed **repolarization** of conduction pathways

INTERVALS AND SEGMENTS

P-R Interval: Time from start of **P wave** to start of **Q wave**

P-R Segment: Time between end of **P wave** and start of **Q wave**

S-T Segment: Time from end of **S wave** to start of **T wave**, indicates completed ventricular activation

Q-T Interval: Time from start of **Q wave** to end of **T wave**, reflects total ventricular activity

S-T Interval: Time from end of **S wave** to end of **T wave**

ABNORMALITIES

1° Heart Block: A conduction delay in the heart

2° Heart Block: Partial blockage of electrical signals in the heart

Wolf-Parkinson-White Syndrome: A condition causing rapid heartbeats due to an extra conduction pathway

Lown-Ganong-Levine Syndrome: A disorder with short P-R interval causing abnormal heart rhythms

Myocardial Infarction: A **heart attack** due to blocked blood flow

Ischemia: Reduced blood supply to the heart

CLINICAL APPLICATIONS

Arrhythmias: Irregular heartbeats

Electrolyte Imbalances: Abnormal levels of minerals affecting heart function

Myocarditis: Inflammation of heart muscles

Cardiomyopathies: Diseases of the heart muscle

Endocrine Disorders: Hormonal imbalances affecting the heart

Conduction Defects: Issues in electrical signal transmission

CARDIAC MONITORING

Cardiac Surgery: Operations related to heart conditions

Thoracic Surgery: Procedures involving the chest

Acute Myocardial Infarction: Emergency due to heart attack

Postoperative Intensive Care: Monitoring after surgery

Emergencies: Critical situations requiring continuous ECG tracking

PHYSIOLOGICAL VARIATIONS OF HEART RATE

Vagal Tone: The activity of the vagus nerve that slows down heart rate

Venous Return: The amount of blood returning to the heart

Cardiac Output: The volume of blood the heart pumps per minute

Marey's Reflex: A reflex where low blood pressure due to standing causes an increase in heart rate

Sinus Arrhythmia: A natural variation in heart rate during breathing

Barometric Pressure: The pressure exerted by the atmosphere

Bradycardia: A slow heart rate

Asystole: Complete absence of heart activity

CARDIOVASCULAR REFLEXES

Baroreflex: A reflex that helps regulate blood pressure by adjusting heart rate

Baroreceptors: Sensors in blood vessels that detect changes in blood pressure

Nucleus Tractus Solitarius (NTS): A brain region that processes signals from baroreceptors

Nucleus Ambiguus: A brain region that controls vagal activity and heart rate

Chemoreflex: A reflex that adjusts heart rate based on blood oxygen and carbon dioxide levels

Chemoreceptors: Sensors that detect oxygen, carbon dioxide, and pH levels in the blood

Inspiratory Center: The brain area that controls breathing

HORMONAL REGULATION OF HEART RATE

Adrenaline: A hormone that increases heart rate by stimulating beta receptors

Beta Receptors: Proteins on heart cells that respond to adrenaline and noradrenaline

Cyclic AMP (cAMP): A molecule that helps transmit adrenaline's effects in cells

Noradrenaline: A hormone that constricts blood vessels, increasing blood pressure

Reflex Bradycardia: A decrease in heart rate due to a rise in blood pressure

Vasopressin: A hormone that increases blood pressure and can lower heart rate reflexively

Thyroxin: A hormone that regulates metabolism and heart rate

Hypothyroidism: A condition where low thyroid hormone levels cause a slow heart rate

Hyperthyroidism: A condition where high thyroid hormone levels cause a fast heart rate

CARDIAC OUTPUT (CO)

Cardiac Output: The volume of blood pumped by each ventricle per minute

Stroke Volume: The amount of blood ejected by each ventricle per heartbeat

REGULATION OF CARDIAC OUTPUT

Peripheral Resistance: The resistance blood encounters in the blood vessels

End Diastolic Volume (EDV): The volume of blood in the ventricles at the end of filling (diastole)

Chronotropic: Refers to factors that affect heart rate

Inotropic: Refers to factors that affect the strength of heart muscle contractions

STARLING'S LAW

Frank-Starling Phenomenon: The relationship where increased preload (EDV) leads to stronger heart contractions

MYOCARDIAL CONTRACTILITY (INOTROPIC EFFECT)

Intrinsic Factors: Factors originating from within the heart that affect contractility

Post-Extrasystolic Potentiation: Increased strength of the heartbeat following an extra heartbeat due to calcium buildup

Force-Frequency Relation: The relationship where an increase in heart rate leads to increased contractility

Extrinsic Factors: Factors originating from outside the heart that influence contractility

NEURAL FACTORS

Norepinephrine: A neurotransmitter that increases heart rate and contractility

Acetylcholine (ACh): A neurotransmitter that reduces heart rate and contractility

HORMONAL FACTORS

Epinephrine: A hormone that increases heart rate and contractility

Glucagon: A hormone that enhances contractility by increasing cyclic AMP

Thyroxin: A hormone that boosts metabolism and increases contractility

Insulin: A hormone that enhances heart muscle contractility

Cortisol: A hormone that enhances contractility and works with catecholamines

ION EFFECTS

Intracellular Ca^{++} : Calcium ions inside cells that are crucial for heart muscle contraction

Na^+ and K^+ : Sodium and potassium ions that can decrease myocardial contractility

DRUGS

Beta-blockers: Medications that reduce heart rate and contractility

Calcium Channel Blockers: Medications that can increase heart contractility

Theophylline and Caffeine: Stimulants that enhance myocardial contractility

Digitalis: A medication that increases contractility by inhibiting Na-K ATPase

Quinidine: A drug that decreases contractility by extending the refractory period

Barbiturates: Drugs that decrease excitability and contractility of the heart

OTHER FACTORS

Venous Return: The volume of blood returning to the heart, influencing EDV

Ventricular Compliance: The ability of the ventricles to stretch and accommodate blood volume

Diastolic Pause: The resting phase of the heart that allows for filling before contraction

Atrial Systole: The contraction of the atria contributing to ventricular filling

HEART RATE (CHRONOTROPIC EFFECT)

Chronotropic Effect: The effect that influences the rate of heartbeat

Staircase Phenomenon: The phenomenon where increased heart rate leads to stronger contractions due to increased calcium levels

Venous Return (VR): The volume of blood returning to the heart from the veins

PERIPHERAL RESISTANCE

Arteriolar Constriction: Narrowing of the small arteries, increasing resistance and affecting blood flow

Aortic Pressure: The pressure in the aorta, influencing cardiac output

Anrep Effect: The phenomenon where increased afterload leads to an increase in contractility to maintain cardiac output

ARTERIAL BLOOD PRESSURE

Arterial Blood Pressure: The lateral force exerted by blood on the walls of arteries

Systolic Blood Pressure: The pressure in the arteries during the contraction of the heart (120 mmHg)

Diastolic Blood Pressure: The pressure in the arteries when the heart is at rest between beats (80 mmHg)

PHYSIOLOGICAL VARIATIONS IN ARTERIAL BLOOD PRESSURE

Diurnal: Relating to the daily cycle of biological activity; blood pressure follows a circadian rhythm

Circadian Rhythm: The physical, mental, and behavioral changes following a 24-hour cycle

AGE

Systolic BP: The upper number in blood pressure readings, representing the maximum pressure during heartbeats

SEX

Sex Hormones: Hormones that influence physical and physiological differences between males and females

SURFACE AREA

Surface Area: The total area of the surface of an object; in this context, it relates to body size and blood pressure

DIGESTION

Sympathetic Stimulation: Activation of the sympathetic nervous system, which can increase heart rate and blood pressure

Vasodilatation: The widening of blood vessels, which can lower blood pressure in specific areas (like the gastrointestinal tract)

SLEEP

Metabolic Rate: The rate at which the body uses energy; it decreases during sleep

Cortical Stimulation: Activation of the brain's cortex, which can influence blood pressure

Vasomotor Center (VMC): A region in the brain that regulates blood vessel diameter and blood pressure

POSTURE

Posture: The position of the body; changes in posture can affect blood pressure readings

REFLEX MECHANISMS IN BP REGULATION

Blood Pressure (BP): The pressure exerted by circulating blood on the walls of blood vessels.

BARORECEPTOR REFLEXES

Baroreceptors: Mechanoreceptors that detect changes in blood pressure by sensing the stretch of blood vessel walls.

Carotid Sinus Reflex: A reflex mechanism involving baroreceptors located in the carotid sinus, which helps regulate blood pressure.

Aortic Arch Reflex: A reflex mechanism involving baroreceptors located in the aortic arch, contributing to blood pressure regulation.

Cardioinhibitory Centre (CIC): A part of the brain that decreases heart rate when activated.

Nucleus Tractus Solitarius (NTS): A region in the brainstem that processes sensory information, including blood pressure signals.

Vasomotor Centre (VMC): A part of the brain that regulates blood vessel constriction and relaxation, influencing blood pressure.

Vagal Tone: The level of activity of the vagus nerve, which decreases heart rate and promotes relaxation.

Sympathetic Tone: The level of activity of the sympathetic nervous system, which increases heart rate and blood pressure.

BARORECEPTOR REFLEX IN RESPONSE TO INCREASED BP

Bradycardia: A slower than normal heart rate.

BARORECEPTOR REFLEX IN RESPONSE TO DECREASED BP

Tachycardia: A faster than normal heart rate.

Peripheral Resistance: The resistance of the arteries to blood flow, influencing blood pressure.

Myocardial Contractility: The strength of the heart's contraction, affecting the amount of blood pumped with each heartbeat.

MAREY'S REFLEX (BARORECEPTOR REFLEX)

Marey's Law: The principle that heart rate decreases as blood pressure increases, indicating an inverse relationship.

Short-Term Regulatory Mechanism: A physiological response that quickly adjusts to changes in blood pressure over a brief time.

Hypertension: A condition characterized by abnormally high blood pressure.

SYSTOLIC AND DIASTOLIC BP

Systolic BP (SBP): The maximum pressure in the arteries when the heart contracts during systole.

Diastolic BP (DBP): The minimum pressure in the arteries at the end of ventricular diastole, when the heart is at rest.

Pulse Pressure (PP): The difference between systolic and diastolic pressure, indicating the force that the heart generates each time it beats.

FACTORS MAINTAINING BP

Cardiac Output (CO): The total volume of blood ejected by the heart per minute, influencing blood pressure.

Circulating Blood Volume: The amount of blood present in the circulatory system, necessary to maintain adequate pressure.

Elasticity of Vessel Walls: The ability of blood vessel walls to stretch and recoil, affecting blood pressure during heart contractions and relaxations.

Viscosity of Blood: The thickness of blood, affecting its flow and the resistance encountered in the blood vessels.

Peripheral Vascular Resistance (PR): The resistance that blood faces as it flows through blood vessels, primarily influenced by vessel diameter and length.

CARDIAC OUTPUT

Stroke Volume: The volume of blood pumped from the heart with each beat, contributing to cardiac output.

CIRCULATING BLOOD VOLUME

Haemorrhage: Excessive blood loss leading to a decrease in blood volume and blood pressure.

Cushing's Syndrome: A condition causing excessive cortisol levels, leading to increased blood volume and pressure.

ELASTICITY OF THE VESSEL WALL

Distension: The ability of blood vessels to stretch in response to increased pressure during systole.

Recoil: The ability of blood vessels to return to their original shape after being stretched, maintaining diastolic pressure.

Arteriosclerosis: The thickening and hardening of the arterial walls, leading to decreased elasticity.

VISCOSITY OF BLOOD

Polycythaemia: An increase in red blood cell mass, leading to higher blood viscosity.

Acidosis: A condition characterized by increased acidity in the body fluids, affecting blood viscosity.

Dehydration: A state of reduced body fluids, which can concentrate blood and increase viscosity.

Anaemia: A condition characterized by a lower than normal number of red blood cells, leading to decreased blood viscosity.

Nephritis: Inflammation of the kidneys that can affect blood composition and viscosity.

PERIPHERAL VASCULAR RESISTANCE

Velocity of Blood: The speed at which blood flows through the vessels, influencing resistance and blood pressure.

Elasticity of Vessel Walls: The capacity of the blood vessels to change shape with varying blood pressure.

Lumen of the Vessel: The internal space within a blood vessel through which blood flows; a larger lumen reduces resistance.

Extravascular Compression: External pressure applied to blood vessels, affecting their ability to conduct blood.

Beri Beri: A disease caused by thiamine deficiency, leading to peripheral neuropathy and vasodilation.

Hyperthyroidism: A condition where the thyroid gland is overactive, causing an increase in metabolism and vasodilation.

FACTORS INCREASING PERIPHERAL RESISTANCE

Old Age: Natural physiological changes leading to reduced vessel elasticity and increased resistance.

Atherosclerosis: The buildup of fatty deposits in the arteries, causing narrowing and increased resistance.

FACTORS DECREASING PERIPHERAL RESISTANCE

Vasodilatation: The widening of blood vessels, reducing resistance and lowering blood pressure.

HEART SOUNDS

Definition: Noises generated by the heart during the cardiac cycle.

1. FIRST HEART SOUND (S1)

- **Production:** Closure of the mitral and tricuspid valves.
- **Clinical Significance:** Indicates the start of systole; loudness or softness indicates various heart conditions.

2. SECOND HEART SOUND (S2)

- **Production:** Closure of the aortic and pulmonary valves.
- **Clinical Significance:** Marks the start of diastole; splitting can indicate normal physiological changes or heart issues.

3. THIRD HEART SOUND (S3)

- **Production:** Rapid ventricular filling.
- **Clinical Significance:** May be normal in young people or indicate heart failure in adults.

4. FOURTH HEART SOUND (S4)

- **Production:** Atrial contraction against a stiff ventricle.
- **Clinical Significance:** Indicates ventricular hypertrophy or decreased compliance.

5. MURMURS

- **Definition:** Abnormal sounds caused by turbulent blood flow.
- **Clinical Significance:** Indicates potential valvular diseases.